Basics & Reasoning Error Analysis TEACHER KEY

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Segment addition

ORIGINAL PROBLEM

B is between A and C. AB = 3x + 2, BC = 2x − 1, and AC = 36. Find AB.

INCORRECT WORKED SOLUTION

$3x + 2 = 2x - 1$

$x = -3$

A. EXPLAIN THE MISTAKE (1 POINT)

The segment parts must ADD to the whole: AB + BC = AC. They are not equal to each other (that would only be true if B were a midpoint).

B. SHOW CORRECT WORK (1 POINT)

$AB + BC = AC$

$(3x + 2) + (2x - 1) = 36$

$5x + 1 = 36$

$x = 7$

$AB = 3(7) + 2 = 23$

2. Midpoint

ORIGINAL PROBLEM

M is the midpoint of AB, and AB = 26. If AM = 4x − 3, find x.

INCORRECT WORKED SOLUTION

$AM = AB$

$4x - 3 = 26$

$x = \frac{29}{4}$

A. EXPLAIN THE MISTAKE (1 POINT)

A midpoint splits the segment into two EQUAL parts, so AM is HALF of AB: AM = 13.

B. SHOW CORRECT WORK (1 POINT)

$AM = \frac{1}{2} \cdot AB = 13$

$4x - 3 = 13$

$4x = 16$

$x = 4$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Basics & Reasoning Error Analysis TEACHER KEY

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

3. Complementary angles

ORIGINAL PROBLEM

∠A and ∠B are complementary. m∠A = (3x + 10)° and m∠B = (2x)°. Find x.

INCORRECT WORKED SOLUTION

3x + 10 + 2x = 180

5x + 10 = 180

5x = 170

x = 34

A. EXPLAIN THE MISTAKE (1 POINT)

Complementary angles add to 90°. A sum of 180° is for SUPPLEMENTARY angles.

B. SHOW CORRECT WORK (1 POINT)

3x + 10 + 2x = 90

5x + 10 = 90

5x = 80

x = 16

4. Linear pair

ORIGINAL PROBLEM

Two angles form a linear pair and measure (5x + 10)° and (3x + 10)°. Find x.

INCORRECT WORKED SOLUTION

5x + 10 = 3x + 10

2x = 0

x = 0

A. EXPLAIN THE MISTAKE (1 POINT)

The angles in a linear pair are SUPPLEMENTARY: they add to 180°. They are not necessarily equal. (With x = 0, the angles would only total 20°.)

B. SHOW CORRECT WORK (1 POINT)

(5x + 10) + (3x + 10) = 180

8x + 20 = 180

8x = 160

x = 20

Angles: 110° and 70°

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Basics & Reasoning Error Analysis TEACHER KEY

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

5. Converse

ORIGINAL PROBLEM

Write the converse of: "If a figure is a square, then it is a rectangle."

INCORRECT WORKED SOLUTION

Converse: If a figure is not a square, then it is not a rectangle.

A. EXPLAIN THE MISTAKE (1 POINT)

That statement is the INVERSE (both parts negated). The converse just SWAPS the hypothesis and conclusion, with no negating.

B. SHOW CORRECT WORK (1 POINT)

Swap the two parts:

Converse: If a figure is a rectangle, then it is a square.

(False: a 2 × 5 rectangle is a counterexample.)

6. Contrapositive

ORIGINAL PROBLEM

Write the contrapositive of: "If it rains, then the game is canceled."

INCORRECT WORKED SOLUTION

Contrapositive: If it does not rain, then the game is not canceled.

A. EXPLAIN THE MISTAKE (1 POINT)

That statement is the INVERSE. The contrapositive SWAPS and NEGATES: "If the game is not canceled, then it does not rain."

B. SHOW CORRECT WORK (1 POINT)

Swap the parts, then negate both:

Contrapositive: If the game is not canceled, then it does not rain.

Basics & Reasoning Error Analysis TEACHER KEY

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

7. Counterexample

ORIGINAL PROBLEM

Is the statement "All prime numbers are odd" true or false?

INCORRECT WORKED SOLUTION

3 is prime and odd.

5 is prime and odd.

7 is prime and odd.

The statement is true.

A. EXPLAIN THE MISTAKE (1 POINT)

Examples that agree can never prove a statement for ALL cases, but one counterexample disproves it. The prime number 2 is even.

B. SHOW CORRECT WORK (1 POINT)

Look for a prime number that is not odd.

2 is prime and even.

The statement is false (counterexample: 2).

8. Inductive vs. deductive

ORIGINAL PROBLEM

Test: 1 + 3 = 4, 3 + 5 = 8, 5 + 7 = 12 (all even). Is "the sum of two odd numbers is even" proven?

INCORRECT WORKED SOLUTION

Three examples all gave even sums.

So it is proven for all odd numbers.

A. EXPLAIN THE MISTAKE (1 POINT)

Checking examples is INDUCTIVE reasoning. It supports a conjecture but does not prove it. A proof needs a general argument.

B. SHOW CORRECT WORK (1 POINT)

Examples give a conjecture, not a proof.

Odd numbers have the form 2m + 1 and 2n + 1.

(2m + 1) + (2n + 1) = 2(m + n + 1), which is even.

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Basics & Reasoning Error Analysis TEACHER KEY

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

9. Law of detachment

ORIGINAL PROBLEM

Given: If a number is divisible by 6, then it is divisible by 3. The number 15 is divisible by 3. What can you conclude?

INCORRECT WORKED SOLUTION

If divisible by 6, then divisible by 3.

15 is divisible by 3.

So 15 is divisible by 6.

A. EXPLAIN THE MISTAKE (1 POINT)

This reverses the conditional. The Law of Detachment needs the HYPOTHESIS to be true. Here only the conclusion is true. (15 is NOT divisible by 6.)

B. SHOW CORRECT WORK (1 POINT)

"Divisible by 3" is the conclusion, not the hypothesis.

The Law of Detachment does not apply.

No valid conclusion. (15 is not divisible by 6.)

10. Biconditional

ORIGINAL PROBLEM

Write as a biconditional, if possible: "If $x = 5$, then $x^2 = 25$."

INCORRECT WORKED SOLUTION

$x = 5$ if and only if $x^2 = 25$.

A. EXPLAIN THE MISTAKE (1 POINT)

A biconditional is only valid when BOTH the conditional and its converse are true. The converse is false: $x = -5$ also gives $x^2 = 25$.

B. SHOW CORRECT WORK (1 POINT)

Converse: If $x^2 = 25$, then $x = 5$.

False: $x = -5$ is a counterexample.

No biconditional can be written.